

Kumar Aryan

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EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering

Bhopal, India

2023 – 2027

EXPERIENCE

Amazon ML Summer School

Amazon

Remote

August 2025

- **Implemented** supervised, unsupervised, and complex **machine learning algorithms** using a suite of **Python libraries** such as scikit-learn, NumPy, and pandas
- Developed machine learning models for diverse projects using advanced algorithms and tools, **achieving accuracy rates exceeding 95%**

PROJECTS

SkillMapper | *React.js, Node.js, Express, MongoDB Atlas, JWT, Langchain*

[Live Link](#)

- Developed a **MERN** stack web app that generates personalized learning roadmaps based on user's skills, goals and time availability, with static alternatives for beginners
- Integrated **LangChain** with **Gemini API** to dynamically create structured, role-specific learning paths using AI, tailored to each user's profile
- Implemented secure authentication using **JWT** and **protected routes** to manage user sessions and access control
- Built a dynamic progress tracking system that allows users to mark completed topics and capstone projects, with persistent storage in **MongoDB**

PitchDeck | *React.js, Node.js, Express, MongoDB Atlas, Multer*

[GitHub](#)

- Developed a full-stack web application using **Node.js** and **Express.js** for the backend, built **REST APIs** with **React.js** as the frontend
- Implemented JWT-based authentication for secure user login and role-based access, enabling investors, startups, and contributors to interact within the platform
- Implemented **database-driven filtering** to match startups with investors based on industry, funding stage, and business model
- Integrated **Cloudinary** and **Multer** for secure and efficient storage and uploads of pitch videos

Genomic Admixture Modeling Pipeline | *Python, admix tools, AADR, qpAdm*

[GitHub](#)

- Built a **first-of-its-kind** scalable qpAdm wrapper, enabling a fully reproducible admixture pipeline via a unified Jupyter Notebook, removing Linux/R setup overhead
- Performed high-dimensional SNP preprocessing and validation on 124K markers across 1000+ individuals using the **AA DR dataset** provided by **David Reich Labs** with Python (NumPy, Pandas)
- Applied qpAdm, PCA, and f-statistics to analyze Indian population structure, modeling key ancestries (**Russia Andronovo, ILA, Iran Shar-i-Sokhta**) with automated batch execution

TECHNICAL SKILLS

Languages: Java (DSA), JavaScript, Python

Frameworks: React.js, Express.js, Tailwind CSS, LangChain, Docker

Tools & Platforms: Amazon AWS, Git, GitHub, Power BI, MongoDB, Postman, Render

Key Concepts: DSA, JWT Auth, MVC Architecture, Operating Systems, DBMS, Computer Networks, qpAdm

Soft Skills: Problem Solving, Teamwork, Communication, Creativity, Ownership

ACHIEVEMENTS

- Solved **200+** Data Structures and Algorithms problems on **LeetCode** and **Codeforces**, enhancing problem-solving skills and algorithmic thinking
- Achieved a **Codeforces** contest rating of **1300**, placing in the top 30% of active global participants in competitive programming
- Finalist in **CyberSafety Hackathon** organised by MP Police, securing a spot among the **top 5** teams